

# Computer Science & Information Technology

## Computer Networks

**DPP: 1**

### MAC Sub-Layer

**Q1** Match the following.

| LAN Standard      | Channel Access Protocol |
|-------------------|-------------------------|
| (i) IEEE 802.3    | (p) CSMA/CA             |
| (ii) IEEE 802.5   | (q) Token               |
| (iii) IEEE 802.11 | (r) CSMA/CD             |

- (A) (i) - (p), (ii) - (q), (iii) - (r)  
 (B) (i) - (r), (ii) - (q), (iii) - (p)  
 (C) (i) - (r), (ii) - (p), (iii) - (q)  
 (D) (i) - (q), (ii) - (r), (iii) - (p)

**Q2** Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 10,000 bits. The channel transmission rate is 2 Mbps. Calculate vulnerable time (in millisecond) ?

**Q3** Consider a network using the pure ALOHA medium access control protocol, where each frame transmission delay is 2 millisecond. The aggregate number of transmissions across all the nodes is 6000 frames per second. Calculate mean number of frame transmissions per frame time ?

**Q4** Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 4 Mbps. The aggregate number of transmissions across all the nodes is 2,000 frames per second. The average number

of frames successfully transmitted per second is \_\_\_\_\_ (rounded to the nearest integer).

**Q5** Consider a network using the slotted ALOHA medium access control protocol, where each frame is of length 4,000 bits. The channel transmission rate is 1 Mbps. Calculate vulnerable time (in millisecond) ?

**Q6** Consider a network using the slotted ALOHA medium access control protocol, where each frame is of length 2,000 bits. The channel transmission rate is 4 Mbps. The aggregate number of transmissions across all the nodes is 1,000 frames per second. The average number of frames successfully transmitted per second is \_\_\_\_\_. (rounded to the nearest integer)

**Q7** Consider a slotted LAN with four number of hosts. Each hosts have data to transmit and transmit with probability 0.2 in every slots. What is the probability (upto three decimal places) that a particular host can transmit in a slot without any collision ?

**Q8** Consider a slotted LAN with 30 number of nodes. Each nodes have data to transmit and transmit with probability P in every slots. What is the probability that any node can transmit in a slot without any collision ?

- (A)  $29 * P * (1-P)^{28}$   
 (B)  $29 * P * (1-P)^{29}$   
 (C)  $30 * P * (1-P)^{29}$   
 (D)  $30 * P * (1-P)^{30}$



- Q9** Consider a slotted LAN with four number of hosts. The hosts have data to transmit and transmit in every slots with the probabilities 0.2, 0.3, 0.4 and 0.5 respectively. What is the probability (upto three decimal places) that any host can transmit in a slot without any collision ?
- Q10** Which is NOT a multiple access protocol ?  
(A) ALOHA (B) CSMA  
(C) CRC (D) Token
- Q11** The minimum frame size (in bytes) required for a CSMA/CD based computer network running at 10 Mbps on a 4 Km cable with a link speed of  $2 \times 10^7$  meter per second is :
- Q12** Determine the maximum length (in kilo-meter) of the cable for transmitting data at a rate of 10 Mbps in an Ethernet LAN with frames of size 2000 bytes. Assume the signal speed in the cable to be  $10^7$  meter per second.
- Q13** Consider A and B are the only two stations on an Ethernet LAN. Each has a steady queue of frames to transmit. Both A and B attempt to transmit a frame, collide and A wins first back-off race. What is the probability (up to three decimal places) that B wins the second back-off race ?



## Answer Key

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**Q1 (B)**

**Q2 10~10**

**Q3 12~12**

**Q4 367~371**

**Q5 4~4**

**Q6 303~305**

**Q7 0.102~0.103**

**Q8 (C)**

**Q9 0.394~0.394**

**Q10 (C)**

**Q11 5000~5000**

**Q12 0.8~0.8**

**Q13 0.125~0.125**



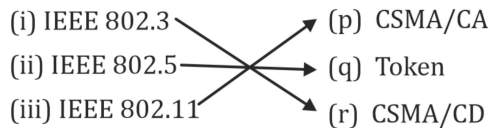
[Android App](#) | [iOS App](#) | [PW Website](#)

# Hints & Solutions

## Q1 Text Solution:

LAN Standard

Channel Access Protocol



## Q2 Text Solution:

$$t_x = \frac{\text{Frame size}}{\text{Bandwidth}} = \frac{10000 \text{ bits}}{2 \text{ Mbps}} = \frac{10 \times 10^3 \text{ bits}}{2 \times 10^6 \text{ bits/sec.}}$$

$$= 5 \text{ ms}$$

Pure ALOHA:

$$\text{Vulnerable time} = 2t_x = 10 \text{ ms}$$

## Q3 Text Solution:

$$t_x = 2 \text{ ms}$$

$$1 \text{ sec.} \rightarrow 6000 \text{ frames}$$

$$10^3 \text{ ms} \rightarrow 6 \times 10^3 \text{ frames}$$

$$1 \text{ ms} \rightarrow 6 \text{ frames}$$

$$(t_x = 2 \text{ ms}) \rightarrow 12 \text{ frames}$$

## Q4 Text Solution:

$$t_x = \frac{\text{Frame size}}{\text{Bandwidth}} = \frac{1000 \text{ bits}}{4 \text{ Mbps}} = \frac{10^3 \text{ bits}}{4 \times 10^6 \text{ bits/sec.}}$$

$$t_x = \frac{1}{4 \times 10^3} \text{ sec.}$$

$$1 \text{ sec.} \rightarrow 2000 \text{ frames}$$

$$\left(t_x = \frac{1}{4 \times 10^3} \text{ sec.}\right) \Rightarrow \left(\frac{2000}{4 \times 10^3}\right) \text{ frames}$$

$$\Rightarrow \frac{1}{2} \text{ frame}$$

$$\text{Mean number of frames transmitted per frame time (G)} = \frac{1}{2}$$

$$\text{Throughput (S)} = ?$$

Pure ALOHA:

$$S = G \times e^{-2G}$$

$$\Rightarrow \frac{1}{2e} = 0.1893$$

Average number of frames successfully transmitted per sec.

$$1 \text{ sec.} \rightarrow (2000 \times S) \text{ frames}$$

$$\rightarrow 367.87 \text{ frames}$$

## Q5 Text Solution:

$$t_x = \frac{\text{Frame size}}{\text{Bandwidth}} = \frac{4000 \text{ bits}}{1 \text{ Mbps}} = \frac{4 \times 10^3 \text{ bits}}{10^6 \text{ bits/sec.}}$$

$$= 4 \text{ ms}$$

Slotted ALOHA:-

$$\text{Vulnerable time} = t_x = 4 \text{ ms}$$

## Q6 Text Solution:

$$t_x = \frac{\text{Frame size}}{\text{Bandwidth}} = \frac{2000 \text{ bits}}{4 \text{ Mbps}} = \frac{2 \times 10^3 \text{ bits}}{4 \times 10^6 \text{ bits/sec.}}$$

$$= \frac{1}{2 \times 10^3} \text{ sec.}$$

$$1 \text{ sec} \rightarrow 1000 \text{ frames}$$

$$\left(t_x = \frac{1}{2 \times 10^3} \text{ sec.}\right) \Rightarrow \left(\frac{1000}{2 \times 10^3}\right) \text{ frames}$$

$$\Rightarrow \frac{1}{2} \text{ frame}$$

Mean number of transmission per frame time (G)

$$= \frac{1}{2}$$

Throughput (S) = ?

Slotted ALOHA:

$$S = G \times e^{-G}$$

$$\Rightarrow \frac{1}{2 \times e^{\frac{1}{2}}} = 0.3032$$

Average number of frame transmitted successfully per sec.

$$1 \text{ sec} \rightarrow (1000 \times S) \text{ frames}$$

$$\rightarrow 303.26 \text{ frames}$$

## Q7 Text Solution:

$$\text{Number of nodes (N)} = 4$$

$$\text{Probability (P)} = 0.2$$

$$\text{Throughput of host} = P \times (1-P)^{(N-1)}$$

$$= (0.2) \times (0.8)^{(4-1)}$$

$$= 0.1024$$

## Q8 Text Solution:

$$\text{Number of nodes (N)} = 30$$

$$\text{Probability} = P$$

$$\text{Throughput of channel} = N \times P \times (1-P)^{(N-1)}$$

$$= 30 \times P \times (1-P)^{29}$$

## Q9 Text Solution:



Android App | iOS App | PW Website

Throughput of channel =  $[0.2 \times 0.7 \times 0.6 \times 0.5] + [0.8 \times 0.3 \times 0.6 \times 0.5] + [0.8 \times 0.7 \times 0.4 \times 0.5] + [0.8 \times 0.7 \times 0.6 \times 0.5] = 0.394$

**Q10 Text Solution:**

- CRC

**Q11 Text Solution:**

**CSMA /CD:**  $t_x \geq 2t_p$

Minimum Frame size =  $(2t_p) \times \text{Bandwidth}$

$$= 2 \times \left( \frac{\text{Distance}}{\text{Signal speed}} \right) \times \text{Bandwidth}$$

$$2 \times \left( \frac{4 \text{ KM}}{20 \times 10^7 \text{ m/sec.}} \right) \times 10 \text{ Mbps}$$

$$= 2 \times \left( \frac{4 \times 10^3}{2 \times 10^7 \text{ m/sec.}} \right) \times 10 \times 10^7 \text{ bit/sec}$$

Minimum frame size =  $4 \times 10^4$  bits

= 5000 bytes

**Q12 Text Solution:**

**Ethernet (CSMA/CD) :**  $t_x \geq 2t_p$

Maximum distance =  $\left( \frac{t_x}{2} \right) \times (\text{Signal speed})$

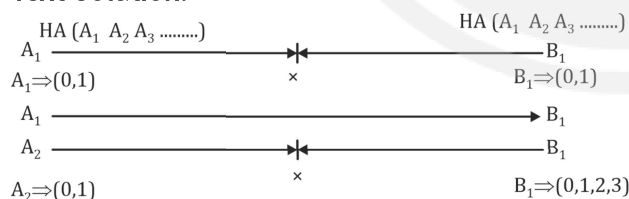
$$= \left( \frac{\text{Frame size} \times \text{Signal speed}}{\text{Bandwidth} \times 2} \right)$$

$$= \left( \frac{2000 \text{ bytes} \times 10^7 \text{ m/sec.}}{10 \text{ Mbps} \times 2} \right)$$

$$\text{Maximum distance} = \frac{2 \times 10^3 \times 8 \text{ bits} \times 10^7 \text{ m/sec.}}{10 \times 10^7 \text{ bits/sec.} \times 2}$$

= 800 meter

= 0.8 Km

**Q13 Text Solution:**

| A <sub>2</sub> | B <sub>1</sub> | Result         |
|----------------|----------------|----------------|
| 0              | 0              | Again collided |
| 1              | 1              |                |
| 0              | 1              | A win          |
| 0              | 2              |                |
| 0              | 3              |                |
| 1              | 2              |                |
| 1              | 3              |                |
| 1              | 0              | B win          |

$= \frac{1}{8} = \boxed{0.125}$

